

Notes: Sloan (The Accounting Review, 1996)

Do Stock Prices Fully Reflect Information in Accruals and Cash Flows About Future Earnings?

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1 Big picture

This paper studies a foundational question in accounting and asset pricing: **do stock prices correctly use information in the accrual and cash-flow components of earnings?** Reported earnings combine relatively objective cash flows with accounting accruals. Sloan asks whether investors understand that these two components have different implications for future earnings.

The main finding is that **earnings generated by accruals are less persistent than earnings generated by cash flows**, but stock prices do not fully reflect this difference. Investors appear to “fixate” on total earnings and underweight the quality of earnings. As a result, firms with high accruals subsequently earn low abnormal returns, while firms with low accruals subsequently earn high abnormal returns.

The paper is the classic source of the **accrual anomaly**: a long-low-accrual, short-high-accrual portfolio earns large abnormal returns, especially in the year after portfolio formation. The pattern is strongest around future earnings announcements, consistent with delayed market recognition of the lower persistence of accrual-based earnings.

2 Incremental contribution of the paper

This paper’s incremental contribution is not simply that earnings predict returns. It shows exactly **which part of earnings** the market misprices.

1. **Decomposes earnings into economically meaningful components.** Prior return-predictability work often used statistical earnings forecasting models. Sloan uses the accounting identity that earnings equal cash flows plus accruals.
2. **Documents different persistence.** The paper shows that the accrual component of earnings is less persistent than the cash-flow component.
3. **Tests market expectations directly.** Using a Mishkin-style framework, the paper compares actual forecasting coefficients with the coefficients implied by stock prices.
4. **Introduces an implementable accrual trading strategy.** Sorting firms on accruals produces economically large abnormal returns, especially in the first year after the accounting information becomes public.
5. **Connects return predictability to earnings-announcement information.** The paper shows that a large fraction of the predictable returns occur around subsequent earnings announcements, consistent with a delayed reaction to accounting information.
6. **Shapes future accounting and finance research.** The paper created a benchmark anomaly linking financial statement analysis, limits to market efficiency, and the pricing of earnings quality.

3 Data and measurement

3.1 Sample

The empirical sample is the intersection of Compustat annual industrial/research files and CRSP monthly stock returns. The core sample contains **40,679 firm-year observations** from 1962–1991. Pre-1962 observations are excluded because of Compustat survivorship concerns and missing data needed to compute accruals. Banks, life insurers, and property/casualty insurers are generally absent because the necessary accrual variables are not available in the same way.

The announcement-period tests use a smaller sample of **16,795 firm-years** from 1973–1991, because quarterly earnings announcement dates are required.

3.2 Earnings, accruals, and cash flows

The paper defines earnings as operating income after depreciation, scaled by average total assets:

$$Earnings_t = \frac{Income\ from\ continuing\ operations_t}{Average\ total\ assets_t}. \quad (1)$$

Operating accruals are computed from balance-sheet and income-statement data:

$$Accruals_t = (\Delta CA_t - \Delta Cash_t) - (\Delta CL_t - \Delta STD_t - \Delta TP_t) - Dep_t, \quad (2)$$

where ΔCA is the change in current assets, $\Delta Cash$ is the change in cash and cash equivalents, ΔCL is the change in current liabilities, ΔSTD is the change in debt included in current liabilities, ΔTP is the change in taxes payable, and Dep is depreciation and amortization expense.

The cash-flow component is measured residually:

$$Cash\ Flows_t = Earnings_t - Accruals_t. \quad (3)$$

All three variables are scaled by average total assets, which makes comparisons across firms and over time meaningful.

3.3 Returns

Annual stock returns begin four months after the fiscal year-end, so financial statements are assumed to be public. The paper uses two abnormal-return measures:

- **Size-adjusted returns:** raw buy-and-hold returns minus returns on a size-matched portfolio.
- **Jensen alphas:** portfolio-level abnormal returns estimated from annual CAPM-style regressions.

4 Models and empirical specifications

4.1 Persistence of total earnings

The first benchmark regression measures the persistence of total earnings:

$$Earnings_{t+1} = \alpha_0 + \alpha_1 Earnings_t + \nu_{t+1}. \quad (4)$$

The coefficient α_1 measures how much current earnings performance carries over to next year.

4.2 Persistence of accruals and cash flows

The key forecasting regression separates earnings into accrual and cash-flow components:

$$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 Cash\ Flows_t + \nu_{t+1}. \quad (5)$$

The central prediction is

$$\gamma_1 < \gamma_2, \quad (6)$$

meaning that accruals are less persistent than cash flows.

4.3 Market efficiency tests

The paper applies the Mishkin framework. The idea is to estimate both a forecasting equation and a stock-return equation. For total earnings, the stock-return equation is

$$Abnormal\ Return_{t+1} = \beta (Earnings_{t+1} - \alpha_0 - \alpha_1^* Earnings_t) + \varepsilon_{t+1}. \quad (7)$$

Market efficiency requires

$$\alpha_1 = \alpha_1^*. \quad (8)$$

For accruals and cash flows, the system becomes

$$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 Cash\ Flows_t + \nu_{t+1}, \quad (9)$$

$$Abnormal\ Return_{t+1} = \beta (Earnings_{t+1} - \gamma_0 - \gamma_1^* Accruals_t - \gamma_2^* Cash\ Flows_t) + \varepsilon_{t+1}. \quad (10)$$

Market efficiency requires

$$\gamma_1 = \gamma_1^* \quad \text{and} \quad \gamma_2 = \gamma_2^*. \quad (11)$$

If investors overestimate accrual persistence, then $\gamma_1^* > \gamma_1$. If investors underestimate cash-flow persistence, then $\gamma_2^* < \gamma_2$.

4.4 Trading strategy

The trading strategy sorts firms annually into accrual deciles. The hedge portfolio takes a long position in the lowest accrual decile and a short position in the highest accrual decile.

The economic logic is straightforward:

- low accrual firms have more cash-flow-based earnings, which are more persistent;
- high accrual firms have more accrual-based earnings, which are less persistent;
- if prices fail to recognize this, low accrual firms should outperform high accrual firms later.

5 Identification and empirical design

This is not a causal treatment-effect paper. It is a joint test of accounting-based earnings forecasts and market efficiency. The empirical design uses several layers of evidence.

1. **Forecasting tests:** show that accruals and cash flows contain different information about future earnings.
2. **Pricing tests:** use Mishkin restrictions to test whether the market prices those components rationally.
3. **Portfolio tests:** test whether the apparent mispricing is economically tradable.
4. **Cross-sectional regressions:** verify that accruals predict future returns even after controlling for other known return predictors.
5. **Announcement-window tests:** ask whether predictable returns arrive when future earnings information is revealed.

The main alternative interpretation is risk. Sloan addresses this by showing that the hedge portfolio has little net beta exposure, that returns are stable across many years, and that a substantial portion of the return is concentrated around subsequent earnings announcements.

6 Main results

6.1 Accruals are less persistent than cash flows

The paper finds that earnings are persistent, but the persistence differs sharply by component. In the actual-value regression, the coefficient on accruals is 0.765 and the coefficient on cash flows is 0.855. In the rank regression, the contrast is even larger: 0.565 for accruals versus 0.838 for cash flows.

This means that one dollar of earnings generated through accruals is less likely to persist into future earnings than one dollar of earnings generated through cash flows.

6.2 Stock prices do not fully reflect the difference

When the paper tests whether prices reflect average earnings persistence, market efficiency is not rejected. But when it decomposes earnings into accruals and cash flows, market efficiency is strongly rejected.

The market-implied coefficient on accruals is larger than the actual accrual persistence coefficient, while the market-implied coefficient on cash flows is smaller than the actual cash-flow persistence coefficient. This is the central pricing result: stock prices behave as if investors overvalue accruals and undervalue cash-flow earnings.

6.3 The accrual strategy earns abnormal returns

A hedge portfolio that buys low-accrual firms and shorts high-accrual firms earns about 10.4% in the first year using size-adjusted returns, with a t-statistic above 4. The return is still positive in year two and fades by year three, matching the timing of the earnings persistence reversal.

6.4 Returns are linked to future earnings announcements

A large fraction of the predictable returns occurs around future quarterly earnings announcements. This timing supports the interpretation that returns reflect delayed market recognition of accounting information rather than compensation for bearing conventional risk.

7 Tables and figures

TABLE 1
Mean (median) Values of Selected Characteristics for Ten Portfolios of Firms Formed Annually by Assigning Firms to Deciles Based on the Magnitude of Accruals.
Sample Consists of 40,679 Firm-years between 1962 and 1991^a

	Portfolio Accrual Ranking									
	Lowest	2	3	4	5	6	7	8	9	Highest
<i>Panel A: Components of Earnings</i>										
Accruals	-0.18 (-0.15)	-0.09 (-0.09)	-0.07 (-0.06)	-0.05 (-0.05)	-0.03 (-0.03)	-0.02 (-0.02)	-0.00 (0.00)	0.02 (0.02)	0.05 (0.05)	0.15 (0.12)
Cash Flows	0.22 (0.23)	0.18 (0.18)	0.16 (0.16)	0.15 (0.15)	0.13 (0.13)	0.12 (0.13)	0.12 (0.11)	0.10 (0.10)	0.08 (0.07)	0.00 (0.00)
Earnings	0.07 (0.07)	0.09 (0.09)	0.10 (0.09)	0.11 (0.10)	0.11 (0.10)	0.11 (0.11)	0.12 (0.11)	0.13 (0.12)	0.13 (0.13)	0.15 (0.13)
<i>Panel B: Risk Proxies</i>										
Portfolio Beta	1.25	0.94	0.95	0.86	0.91	0.94	0.93	0.93	1.06	1.23
Size	3.80 (3.54)	4.57 (4.38)	4.93 (4.84)	4.97 (4.93)	4.94 (4.86)	4.90 (4.84)	4.77 (4.65)	4.62 (4.53)	4.32 (4.20)	3.99 (3.86)
<i>Panel C: Components of Accruals</i>										
Current Asset	-0.08 (-0.06)	-0.01 (-0.01)	0.01 (0.01)	0.02 (0.01)	0.03 (0.02)	0.04 (0.03)	0.05 (0.05)	0.07 (0.07)	0.11 (0.10)	0.21 (0.19)
Current Liability	-0.03 (-0.02)	-0.02 (-0.02)	-0.02 (-0.01)	-0.02 (-0.01)	-0.02 (-0.01)	-0.02 (-0.01)	-0.02 (-0.01)	-0.02 (-0.02)	-0.02 (-0.02)	-0.03 (-0.03)
Depreciation Expense	-0.06 (-0.05)	-0.06 (-0.05)	-0.05 (-0.05)	-0.05 (-0.05)	-0.04 (-0.04)	-0.04 (-0.04)	-0.04 (-0.03)	-0.03 (-0.03)	-0.03 (-0.03)	-0.03 (-0.03)

^a The firm characteristics are computed as follows:
 Accruals = the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets.
 Earnings = income from continuing operations divided by average total assets.
 Cash flows = the difference between earnings and accruals (as defined above).
 Portfolio beta = the beta coefficient from a time-series regression of the excess return on the portfolio over the risk-free rate on the excess return on the market over the risk-free rate using the 30 calendar years in the sample.
 Size = the natural log of the market value of common equity (in millions of dollars) measured at fiscal year end.
 Current asset = the change in non-cash current assets divided by average total assets.
 Current liability = minus the change in current liabilities (exclusive of short-term debt and taxes payable) divided by average total assets.
 Depreciation = minus depreciation expense divided by average total assets.

Table 1. Summary statistics by accrual decile.

TABLE 2
Results from Ordinary Least Squares Regressions of Future Earnings Performance on Current Earnings Performance (t-statistics in parentheses)
Sample Consists of 40,679 Firm-years from 1962 to 1991^a

$$Earnings_{t+1} = \alpha_0 + \alpha_1 Earnings_t + v_{t+1}$$

Panel A: Regressions using actual values

	Pooled	Industry Level			
		Mean	Q1	Median	Q3
α_0	0.015 (32.57)**	0.021	0.014	0.019	0.027
α_1	0.841 (303.98)**	0.773	0.708	0.774	0.863

Panel B: Regressions using decile rankings

	Pooled	Industry Level			
		Mean	Q1	Median	Q3
α_0	1.193 (62.37)**	1.300	0.885	1.187	1.454
α_1	0.783 (253.93)**	0.768	0.699	0.772	0.825

^a Earnings is income from continuing operations divided by average total assets.
 **Denotes significance at the 0.01 level using a two-tailed t-test.

Table 2. Earnings persistence regression.

Table 1. Read:

- The lowest accrual decile has mean accruals of -0.18 and mean cash flows of 0.22 ; the highest accrual decile has mean accruals of 0.15 and mean cash flows of 0.00 .
- Earnings are higher in the high-accrual decile, but those earnings are not backed by cash flows.
- Most of the cross-sectional variation in accruals comes from the current asset component.
- The extreme accrual portfolios both have high betas, but a long-low, short-high hedge portfolio has almost no net beta exposure.

Table 1. Economic message: Sorting on accruals separates firms with high cash-flow earnings from firms with high accrual earnings. This is exactly the accounting distinction the paper uses to test whether investors understand earnings quality.

Table 2. Read:

- Total earnings persistence is high but below one: the pooled actual-value coefficient is 0.841 .
- The rank regression gives a similar but slightly lower persistence coefficient of 0.783 .

- Accounting profitability is mean reverting, but only gradually.

Table 2. Economic message: Total earnings are persistent enough that investors rationally care about them. The more subtle question is whether investors understand which part of earnings is driving that persistence.

TABLE 3

Results from Ordinary Least Squares Regressions of Future Earnings Performance on the Accrual and Cash Flow Components of Current Earnings Performance

(t-statistics in parentheses)

Sample Consists of 40,679 Firm-years from 1962 to 1991^a

$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 Cash\ Flows_t + v_{t+1}$

Panel A: Regressions using actual values

	<i>Pooled</i>	<i>Industry Level</i>			
		<i>Mean</i>	<i>Q1</i>	<i>Median</i>	<i>Q3</i>
γ_0	0.011 (24.05)**	0.019	0.009	0.016	0.022
γ_1	0.765 (186.53)**	0.721	0.635	0.703	0.780
γ_2	0.855 (304.56)**	0.781	0.733	0.777	0.873

F-test of $\gamma_1 = \gamma_2$: 614.01^b

Proportion of cases in which $\gamma_1 < \gamma_2$: 86%^c

Panel B: Regressions using decile rankings

	<i>Pooled</i>	<i>Industry Level</i>			
		<i>Mean</i>	<i>Q1</i>	<i>Median</i>	<i>Q3</i>
γ_0	-2.216 (-55.86)**	-1.691	-2.666	-1.938	-1.459
γ_1	0.565 (141.02)**	0.507	0.438	0.520	0.585
γ_2	0.838 (209.34)**	0.794	0.760	0.805	0.884

F-test of $\gamma_1 = \gamma_2$: 4894.24^b

Proportion of cases in which $\gamma_1 = \gamma_2$: 99%^c

^a Accruals is the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets. Earnings is income from continuing operations divided by average total assets. Cash flows is the difference between earnings and accruals (as defined above).

^b Significant at the 0.01 level using an F-test.

^c Significant at the 0.01 level using a sign test.

**Denotes significance at the 0.01 level using a two-tailed t-test.

^a Accruals is the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets. Earnings is income from continuing operations divided by average total assets. Cash flows is the difference between earnings and accruals (as defined above).

^b Significant at the 0.01 level using an F-test.

^c Significant at the 0.01 level using a sign test.

**Denotes significance at the 0.01 level using a two-tailed t-test.

Table 3. Persistence of accrual and cash-flow components.

Table 3. Read:

- In actual values, accrual persistence is 0.765 while cash-flow persistence is 0.855.
- In decile ranks, accrual persistence is 0.565 while cash-flow persistence is 0.838.
- Equality of the two coefficients is rejected strongly.
- In almost all industry-level regressions, accruals are less persistent than cash flows.

Table 3. Economic message: The table establishes the paper's first key ingredient. Earnings quality matters because cash-flow earnings are more durable than accrual earnings.

Figure 1. Read:

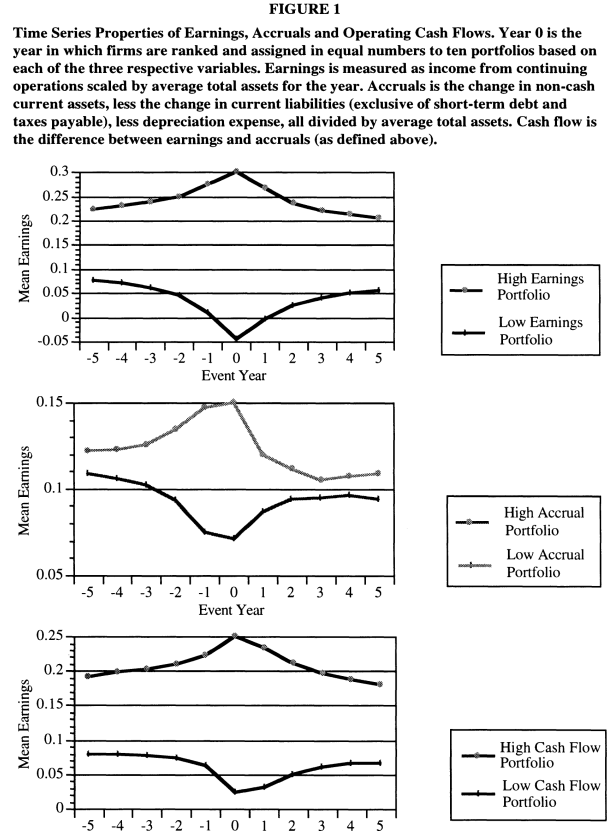


Figure 1. Time-series properties of earnings, accruals, and cash flows.

- Extreme earnings portfolios mean revert slowly.
- Extreme accrual portfolios mean revert quickly, mostly within the first few years.
- Extreme cash-flow portfolios show much more persistent earnings differences.

Figure 1. Economic message: The figure gives the visual intuition behind the accrual anomaly. High accrual earnings fade quickly; high cash-flow earnings persist.

TABLE 4
Results from Nonlinear Generalized Least Squares Estimation of the Stock Price
Reaction to Information in Current Earnings about Future Earnings
Sample Consists of 40,679 Firm-years between 1962 and 1991^a

$$Earnings_{i,t+1} = \alpha_0 + \alpha_1 Earnings_{i,t} + v_{i,t+1}$$

$$Abnormal\ Return_{i,t+1} = \beta(Earnings_{i,t+1} - \alpha_0 - \alpha_1 Earnings_{i,t}) + \varepsilon_{i,t+1}$$

Panel A: Regressions using actual values of financial statement variables

<i>Parameter</i>	<i>Estimate</i>	<i>Asymptotic standard error</i>
α_1	0.841	0.003
α'_1	0.840	0.009
β	1.920	0.032

Test of market efficiency: $\alpha_1 = \alpha'_1$

Likelihood ratio statistic 0.007

Marginal significance level 0.933

Panel B: Regressions using decile rankings of financial statement variables

<i>Parameter</i>	<i>Estimate</i>	<i>Asymptotic standard error</i>
α_1	0.783	0.003
α'_1	0.775	0.009
β	0.082	0.001

Test of market efficiency: $\alpha_1 = \alpha'_1$

Likelihood ratio statistic 0.783

Marginal significance level 0.376

^a The firm characteristics are computed as follows:

Earnings = income from continuing operations divided by average total assets.

Abnormal returns are computed by taking the raw buy-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms.

The decile rankings and decile returns are supplied by CRSP.

The return cumulation period begins four months after the fiscal year-end of the year in which the financial variable are measured.

^a The firm characteristics are computed as follows:
Earnings = income from continuing operations divided by average total assets.
Abnormal returns are computed by taking the raw buy-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms.
The decile rankings and decile returns are supplied by CRSP.
The return cumulation period begins four months after the fiscal year-end of the year in which the financial variables are measured.

Table 4. Market efficiency test using total earnings.

Table 4. Read:

- The actual earnings persistence coefficient is 0.841.
- The market-implied coefficient is 0.840.
- The likelihood ratio test does not reject market efficiency for total earnings.

Table 4. Economic message: Stock prices appear to understand the average persistence of total earnings. The market failure is not about ignoring earnings completely; it is about failing to parse earnings into cash and accrual components.

Table 5. Read:

TABLE 5		
Results from Nonlinear Generalized Least Squares Estimation of the Stock Price Reaction to Information in the Accrual and Cash Flow Components of Current Earnings about Future Earnings		
Sample Consists of 40,679 Firm-years between 1962 and 1991 ^a		
$Earnings_{i,t+1} = \gamma_0 + \gamma_1 Accruals_{i,t} + \gamma_2 Cash\ Flows_{i,t} + v_{i,t+1}$		
$Abnormal\ Return_{i,t+1} = \beta (Earnings_{i,t+1} - \gamma_0 - \gamma_1 Accruals_{i,t} - \gamma_2 Cash\ Flows_{i,t}) + \epsilon_{i,t+1}$		
Panel A: Regressions using actual values of financial statement variables		
Parameter	Estimate	Asymptotic Standard Error
γ_1	0.765	0.004
γ_1'	0.911	0.014
γ_2	0.855	0.003
γ_2'	0.826	0.010
β	1.894	0.032
Test of market efficiency:	$\gamma_1 = \gamma_1'$ and $\gamma_2 = \gamma_2'$	
Likelihood ratio statistic	180.91	
Marginal significance level	0.000	
Panel B: Regressions using decile rankings of financial statement variables		
Parameter	Estimate	Asymptotic Standard Error
γ_1	0.565	0.004
γ_1'	0.675	0.014
γ_2	0.838	0.004
γ_2'	0.747	0.014
β	0.063	0.001
Test of market efficiency:	$\gamma_1 = \gamma_1'$ and $\gamma_2 = \gamma_2'$	
Likelihood ratio statistic	203.75	
Marginal significance level	0.000	

^a The firm characteristics are computed as follows:
Accruals = the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets.
Earnings = income from continuing operations divided by average total assets.
Cash flows = the difference between earnings and accruals (as defined above).
Abnormal returns are computed by taking the raw buy-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms.
The decile rankings and decile returns are supplied by CRSP.
The return calculation period begins four months after the fiscal year-end of the year in which the financial variables are measured.

^a The firm characteristics are computed as follows:
Accruals = the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets.
Earnings = income from continuing operations divided by average total assets.
Cash flows = the difference between earnings and accruals (as defined above).
Abnormal returns are computed by taking the raw buy-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms.
The decile rankings and decile returns are supplied by CRSP.
The return cumulation period begins four months after the fiscal year-end of the year in which the financial variables are measured.

Table 5. Market efficiency test using accruals and cash flows.

- Actual accrual persistence is 0.765, but the market-implied accrual coefficient is 0.911.
- Actual cash-flow persistence is 0.855, but the market-implied cash-flow coefficient is 0.826.
- The market efficiency restrictions are rejected with marginal significance level 0.000.
- The decile-rank version reaches the same conclusion.

Table 5. Economic message: This is the central market-efficiency result. Prices behave as if accruals are too persistent and cash flows are not persistent enough. Investors overprice high-accrual earnings and underprice cash-flow earnings.

Portfolio Accrual Ranking	Size Adjusted Returns ^b			Jensen Alphas ^c		
	year $t+1$	year $t+2$	year $t+3$	year $t+1$	year $t+2$	year $t+3$
Lowest	0.049 (2.65)**	0.016 (1.17)	0.007 (0.55)	0.039 (2.01)*	0.007 (0.40)	0.001 (0.08)
2	0.028 (3.60)**	0.019 (1.65)	0.006 (0.68)	0.020 (1.68)	0.022 (1.53)	0.012 (1.06)
3	0.024 (3.84)**	0.012 (2.27)*	-0.006 (-0.86)	0.018 (1.70)	0.014 (1.28)	-0.006 (-0.72)
4	0.012 (1.66)	0.001 (0.05)	0.020 (2.72)*	0.017 (2.09)*	0.002 (0.17)	0.017 (1.29)
5	0.001 (0.03)	0.002 (0.22)	0.006 (0.86)	0.010 (0.87)	0.004 (0.38)	0.014 (1.12)
6	0.010 (1.43)	0.005 (0.72)	0.016 (1.90)	0.006 (0.57)	0.002 (0.24)	0.003 (0.43)
7	-0.002 (-0.22)	0.003 (0.60)	-0.006 (-0.83)	0.004 (0.39)	0.006 (0.97)	0.005 (0.56)
8	-0.021 (-3.03)**	-0.002 (-0.31)	-0.001 (-0.01)	0.011 (-1.17)	-0.004 (-0.39)	0.002 (0.16)
9	-0.035 (-3.70)**	-0.018 (-2.52)*	-0.015 (-1.60)	-0.028 (-3.04)**	-0.012 (-1.36)	-0.012 (-1.15)
Highest	-0.055 (-3.98)**	-0.032 (-2.25)*	-0.022 (-1.61)	-0.064 (-4.68)**	-0.040 (-2.87)**	-0.036 (-2.47)*
Hedge ^d	0.104 (4.71)**	0.048 (3.15)**	0.029 (1.64)	0.104 (4.42)**	0.048 (2.41)*	0.038 (1.62)

Portfolios are formed annually by assigning firms into deciles based on the magnitude of accruals in year t . The values in parentheses are t -statistics based on the time-series of the annual portfolio abnormal stock returns.

^a Accruals is the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets.

^b The size-adjusted returns are computed by taking the raw buy-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms. The decile rankings and decile returns are supplied by CRSP. The return cumulation period begins four months after the fiscal year-end of the year in which the level of operating accruals is measured. The return cumulation period begins four months after the fiscal year-end.

^c The Jensen alpha is the estimated value of α from $(R_{it} - R_{ft}) = \alpha + \beta_1(R_{mt} - R_{ft}) + \epsilon_{it}$, where R_{ft} denotes the raw buy-hold return to portfolio p in year t , inclusive of dividends and liquidating distributions. R_{ft} is the risk free rate, measured using the contemporaneous annual T-bill yield. R_{mt} is the market return, estimated by cumulating CRSP monthly returns on the equal-weighted NYSE/AMEX index. The return cumulation period begins four months after the fiscal year-end.

^d The hedge portfolio consists of a long position in the lowest accrual portfolio and an offsetting short position in the highest accrual portfolio.

* Denotes significance at the 0.05 level using a two-tailed t -test.

** Denotes significance at the 0.01 level using a two-tailed t -test.

Table 6. Abnormal returns by accrual decile.

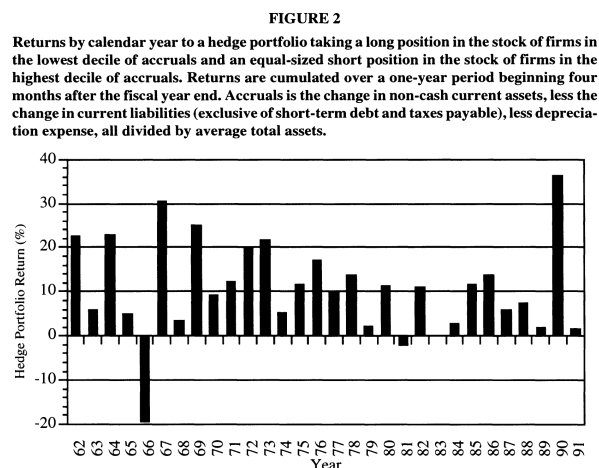


Figure 2. Calendar-year hedge portfolio returns.

Table 6. Read:

- In year $t + 1$, the lowest accrual decile earns a 4.9% size-adjusted abnormal return, while the highest accrual decile earns -5.5%.
- The low-minus-high accrual hedge return is 10.4% in year $t + 1$.
- The hedge return is 4.8% in year $t + 2$ and smaller by year $t + 3$.
- Jensen alpha estimates show a similar pattern.

Table 6. Economic message: The pricing error is economically large. A simple accounting-based trading strategy generates large abnormal returns after the accounting information is publicly available.

Figure 2. Read:

- The annual low-minus-high accrual hedge return is positive in 28 of the 30 years.
- The only negative years are 1966 and 1981.
- The effect is not concentrated in one unusual subperiod.

Figure 2. Economic message: The stability of the hedge returns makes it harder to explain the anomaly as a conventional risk premium or a one-time data artifact.

TABLE 7			
Cross-sectional Regression Tests of the Explanatory Power of Accruals with Respect to Future Annual Stock Returns (t-statistics in parentheses).			
Sample consists of 40,679 firm-years between 1962 and 1991 ^a			
<i>Panel A: Cross-sectional regressions of stock returns on accruals</i>			
	<i>Dependent variable is stock return for:</i>		
	<i>year $t+1$</i>	<i>year $t+2$</i>	<i>year $t+3$</i>
Intercept	0.16 (4.43)**	0.16 (4.42)**	0.17 (4.29)**
Accruals	-0.32 (-4.60)**	-0.16 (-2.75)**	-0.14 (-2.61)*
<i>Panel B: Cross-sectional regressions of stock returns on the components of accruals</i>			
	<i>Dependent variable is stock return for:</i>		
	<i>year $t+1$</i>	<i>year $t+2$</i>	<i>year $t+3$</i>
Intercept	0.17 (4.30)**	0.17 (4.18)**	0.18 (4.28)**
Current asset	-0.35 (-4.81)**	-0.17 (-3.16)**	-0.15 (-2.72)*
Current liability	-0.25 (-3.44)**	-0.08 (-0.98)	-0.04 (-0.49)
Depreciation	-0.14 (-0.57)	-0.01 (-0.05)	-0.02 (-0.07)
Expense			
<i>Panel C: Cross-sectional regressions of stock returns on accruals and other predictors of returns</i>			
	<i>Dependent variable is stock return for:</i>		
	<i>year $t+1$</i>	<i>year $t+2$</i>	<i>year $t+3$</i>
Intercept	0.28 (5.60)**	0.27 (5.42)**	0.22 (4.86)**
Accruals	-0.30 (-6.15)**	-0.10 (-2.19)*	-0.11 (-1.74)
Size	-0.02 (-3.24)**	-0.02 (-2.75)**	-0.01 (-2.08)*
Book-to-market	0.03 (2.31)*	0.02 (1.97)	0.02 (2.32)*
Beta	-0.02 (-0.91)	-0.02 (-0.77)	-0.01 (-0.54)
Earnings-to-price	0.16 (2.04)*	0.05 (0.55)	0.00 (0.01)

Table 7. Fama-MacBeth return regressions, main panels.

Table 7. Read:

- Accruals have negative coefficients for future annual returns, especially in year $t + 1$.
- Decomposing accruals shows that the current asset component is especially important.

TABLE 7 (Continued)	
The number reported are time-series means of the estimated parameters from cross-sectional regressions.	
^a The firm characteristics are computed as follows:	
Accruals	= the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets.
Current asset	= the change in non-cash current assets divided by average total assets.
Current liability	= minus the change in current liabilities (exclusive of short-term debt and taxes payable) divided by average total assets.
Depreciation	= minus depreciation expense divided by average total assets.
Size	= the natural log of the year-end market value of common equity (measured in millions of dollars).
Book-to-market	= the natural log of the ratio of the book value of common equity to the market value of common equity, again measured at the year-end.
Beta	= estimated from a regression of monthly raw returns on the CRSP NYSE/AMEX equal weighted monthly return index. The regression is estimated using the 60-month return period ending four months after each firm's fiscal year-end.
Earnings-to-price	= estimated as earnings-per-share divided by the fiscal-year-end stock price.
Stock returns are inclusive of dividends and any liquidating distributions reported by CRSP.	
The return cumulation period begins four months after the fiscal year-end.	
* Denotes significance at the 0.05 level using a two-tailed t-test.	
** Denotes significance at the 0.01 level using a two-tailed t-test.	

Table 7 continued. Variable definitions and notes.

- The accrual coefficient remains negative after controlling for size, book-to-market, beta, and earnings-to-price.
- The return-prediction evidence is strongest in the first year and weakens over time.

Table 7. Economic message: The accrual anomaly is not simply a disguised version of size, value, beta, or earnings-to-price effects. Accruals add distinct information about future returns.

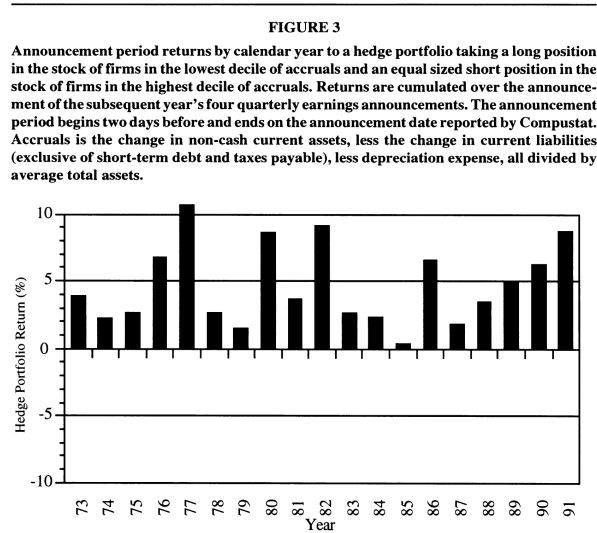


Figure 3. Announcement-period hedge returns.

Figure 3. Read:

- Announcement-period hedge returns are positive in every year in the 1973–1991 announcement sample.
- The returns are measured around the subsequent year's four quarterly earnings announcements.
- The figure suggests that the anomaly is repeatedly revealed when later earnings news arrives.

Figure 3. Economic message: The market does not instantly process the implications of accruals. Instead, prices adjust when future earnings announcements reveal the lower persistence of accrual-heavy earnings.

TABLE 8
Announcement and Non-announcement Period Size-adjusted Portfolio Stock Returns in the Year Following Portfolio Formation (year t+1)
Sample Consists of 16,795 Firm-years between 1973 and 1991^a

Accrual portfolio	Total period return _{t+1} ^c	Announcement period return _{t+1} ^b	Non-announcement period return _{t+1} ^e	Proportion of late reporters _{t+1} ^d
Lowest	0.055 (2.01)*	0.045 (5.45)**	0.009 (0.34)	0.285
2	0.056 (2.04)*	0.027 (3.76)**	0.032 (1.11)	0.276
3	0.010 (0.42)	0.016 (2.49)**	-0.006 (-0.30)	0.272
4	0.012 (0.61)	0.018 (3.37)**	-0.003 (-0.12)	0.285
5	-0.010 (-0.70)	0.014 (3.12)**	-0.023 (-2.00)*	0.272
6	-0.019 (-1.08)	0.016 (3.53)**	-0.034 (-2.14)*	0.259
7	-0.001 (-0.32)	0.012 (2.01)*	-0.014 (-0.76)	0.283
8	-0.013 (-0.71)	0.010 (1.82)	-0.020 (-1.15)	0.279
9	-0.011 (-0.40)	0.010 (1.50)	-0.017 (-0.61)	0.304
Highest	-0.057 (-2.21)*	0.000 (0.02)	-0.051 (-2.04)*	0.345
Hedge ^e	0.112 (6.22)**	0.045 (5.51)**	0.060 (3.41)**	

Portfolios are formed annually by assigning firms into deciles based on the magnitude of accruals in year t. The values in parentheses are t-statistics based on the time-series of portfolio stock returns.

^a Accruals is the change in non-cash current assets, less the change in current liabilities (exclusive of short-term debt and taxes payable), less depreciation expense, all divided by average total assets. The size-adjusted returns are computed by taking the raw buy-and-hold return, inclusive of dividends and any liquidating distributions and subtracting the buy-and-hold return on a size matched, value-weighted portfolio of firms. The size portfolios are based on market-value of equity deciles of NYSE and AMEX firms. The decile rankings and decile returns are supplied by CRSP. The return cumulation period begins four months after the fiscal year-end of the year in which the level of operating accruals is measured.

^b The announcement period return is the cumulative return over the four three-day periods around each of the earnings announcements in the fiscal year following the portfolio formation year. The three day period begins two days prior to the announcement date reported by Compustat and ends on the announcement date reported by Compustat.

^c The non-announcement period return is the cumulative return over the four non-announcement periods following each of the earnings announcements in the fiscal year following the portfolio formation year. The non-announcement period begins on the day after the earnings announcement date reported by Compustat and ends three days prior to the next quarter's earnings announcement date.

^d A firm is classified as a late reporter if any of the four quarterly earnings announcements in the fiscal year following the portfolio formation year are more than two calendar days later than for the corresponding quarter in the portfolio formation year.

^e The hedge portfolio consists of a long position in the lowest accrual portfolio (portfolio 1) and an offsetting short position in the highest accrual portfolio (portfolio 10).

* Denotes significance at the 0.05 level using a two-tailed t-test.

** Denotes significance at the 0.01 level using a two-tailed t-test.

Table 8. Announcement and non-announcement returns.

Table 8. Read:

- In the announcement-date sample, the low-minus-high accrual hedge earns 11.2% in the year after formation.
- About 4.5% occurs during announcement windows and 6.0% outside those windows.
- Over 40% of the predictable return is concentrated in announcement windows, even though these windows represent less than 5% of trading days.
- The low-accrual “good news” side is especially concentrated around earnings announcements.

Table 8. Economic message: This timing evidence links the return anomaly to future earnings realizations. The market correction occurs when the accounting implications become visible in subsequent earnings.

8 Short summary

- **Question:** Do stock prices fully reflect the different future-earnings implications of accruals and cash flows?
- **Key accounting fact:** Accruals are less persistent than cash flows.
- **Key pricing fact:** Prices behave as if investors overestimate accrual persistence and underestimate cash-flow persistence.
- **Trading result:** Low-accrual firms outperform high-accrual firms after portfolio formation.
- **Timing result:** A large share of the predictable returns occurs around later earnings announcements.
- **Interpretation:** Investors fixate on earnings and do not fully process earnings quality until later earnings news arrives.

9 Conclusion

Sloan’s paper shows that financial statement analysis contains information that the stock market does not fully and immediately incorporate. The core accounting insight is simple: earnings backed by cash flows are more persistent than earnings generated by accruals. The core asset-pricing insight is equally powerful: stock prices do not fully reflect this difference.

The paper’s contribution is especially important because it connects a balance-sheet-based accounting measure to future earnings and to future abnormal stock returns. High accruals predict lower earnings persistence and lower subsequent returns. Low accruals predict more persistent earnings and higher subsequent returns. The pattern survives risk adjustments, appears in cross-sectional regressions, and is partly concentrated around future earnings announcements.

The economic interpretation is that investors appear to focus too much on bottom-line earnings and too little on the composition of earnings. This behavior creates predictable mispricing. The paper therefore became a benchmark study in both accounting and finance, shaping later research on earnings quality, investor inattention, limits to arbitrage, and market efficiency.